Rigidity at the minimum local orbit size

Human Compression project: L044 continuation

All marked generators are involutions satisfying string commutations. No intersection property is assumed in any restricted action below.

Lemma 1 (The centralizer of a short anchored action)**.** *Let $T = \langle t_0, \dots, t_{h-1} \rangle$ be transitive on V, where $h \geq 3$, the last mark is nonidentity, and a is fixed by $t_1, \dots, t_{h-1}$. If $|V| \leq h + 2$, the full permutation centralizer of T is trivial.*

Proof. The complete acquired anchored-path lemma gives $|V| \geq h+1$. In degree $h+1$ it identifies $T = \mathrm{Sym}(V)$, whose centralizer is trivial. In degree $h + 2$ it supplies the distinct path $a_0 = a, a_1, \dots, a_h$, with t_i exchanging a_i, a_{i+1} and fixing every earlier path point. Let z be the remaining point.

Suppose $c \neq 1$ centralizes T. A centralizer of a transitive group has no nonidentity element fixing a point. Thus $c(a) \neq a$. It is fixed by all $t_1, \dots, t_{h-1}$. No other path point has this property: a_1 is moved by t_1, and a_k, $k \geq 2$, is moved by t_{k-1}. Consequently $c(a) = z$. In particular all t_i with $i \geq 1$ fix z. The last two marks are now literal overlapping transpositions $(a_{h-1}\ a_h)$ and $(a_{h-2}\ a_{h-1})$: every other point is fixed by their path properties and fixation of z. Their indices are both at least one because $h \geq 3$. Commuting with both transpositions makes c preserve both two-point supports, hence fix their unique common point a_{h-1}. This contradicts semiregularity and proves the lemma. □

Lemma 2 (The larger bound with both factors present)**.** *In the complete two-commuting-anchored-factor setting, assume both active lengths h, u are positive and $h+u \geq 4$. Then the transitive orbit has size at least $h + u + 2$.*

Proof. If neither factor is transitive, the acquired count already gives $2\max(h+1, u+1) \geq h+u+2$. If the left factor is transitive, all right marks except the first fix its anchor and centralize it, so $u = 1$. Thus $h \geq 3$. The right mark gives a nontrivial centralizer element, so the preceding lemma excludes degree at most $h + 2$. The degree is at least $h + 3 = h + u + 2$. Interchange the factors for the other case. The small pairs of lengths $(1,1), (1,2), (2,1)$ are outside this statement. □

Theorem 3 (The minimum orbit and its first omission)**.** *Let $Q = \langle \rho_m, \dots, \rho_y \rangle$ be transitive on W, every displayed restriction nonidentity. Suppose its full local j-omission has exactly two orbits joined by exactly one j-edge. Put $h = j - m$, $u = y - j$ and assume $h, u \geq 2$ and*

$$|W| = y - m + 2 = h + u + 2.$$

Then the two local omission orbits A, B can be named so that $|A| = h + 1$, $|B| = u + 1$ and

$$Q_{<j} = \mathrm{Sym}(A), \qquad Q_{>j} = \mathrm{Sym}(B), \qquad Q = \mathrm{Sym}(W),$$

where each displayed group fixes the complementary set pointwise. If the first omission $Q_{>m}$ is intransitive, then for one point $z \in A$,

$$Q_{>m} = \mathrm{Sym}(W \setminus \{z\}),$$

again with z fixed pointwise.

Proof. Use the actual two local omission orbits and their anchored commuting left and right factors, as in the complete local orbit-size proof. The greatest left and right active lengths are h, u. They cannot occur only on the same side: the preceding lemma would make that side have at least $h + u + 2$ points, leaving no point for the other side. Thus the maxima occur on opposite sides. The anchored bounds are saturated on both: $|A| = h+1$ and $|B| = u+1$. The exhausted anchored path lemma identifies the left image on A as natural symmetric, and the right image on B as natural symmetric. Both degrees are at least three. The opposite commuting factor therefore acts trivially on each side, proving the first two pure subgroup formulas.

Write the unique crossing as $\{a_0, b_0\}$. The permutation ρ_j consists of this transposition and permutations internal to A and B. Those internal permutations belong to the two pure symmetric subgroups, so the generated group contains the single crossing transposition. Together these groups generate $\mathrm{Sym}(W)$.

We now analyze the first omission, rather than assuming its form. On A reverse the left indices: $t_i = \rho_{j-1-i}|_A$ for $0 \le i < h$. The exhausted path is $a_0, a_1, \ldots, a_h$. Its complete proof gives

$$\langle t_i, \ldots, t_{h-1} \rangle = \mathrm{Sym}(\{a_i, \ldots, a_h\}).$$

It also determines the individual marks: each t_i is its path transposition, with only one possible extra transposition. For $i = h - 3$ it may additionally exchange a_{h-1}, a_h, when $h \ge 3$. Indeed the only possible extra action is on $\{a_{i+2}, \ldots, a_h\}$ and centralizes its full symmetric group; that centralizer is nontrivial only when the set has size two.

The first omitted mark is $\rho_m = t_{h-1}$. If the extra transposition just described were present, the remaining left marks would still have a connected orbit graph on A: their ordinary path edges join a_0 through a_{h-1}, and the extra edge joins a_{h-1} to a_h. With the crossing and the full right action this would make $Q_{>m}$ transitive on W, contrary to its hypothesis. Hence no extra is present, and the remaining left marks give the pure $\mathrm{Sym}(A \setminus \{a_h\})$ while fixing a_h.

For $h \ge 3$, ρ_j centralizes the pure $\mathrm{Sym}(\{a_1, \ldots, a_h\}) = Q_{\le j-2}$, so fixes its entire support. Here the equality refers to the left marks $m, \ldots, j-2$ inside Q. For $h = 2$, its only possible additional action on A is $(a_1\ a_2)$. If that were present, the surviving left edge $(a_0\ a_1)$, the crossing and this extra edge would likewise make $Q_{>m}$ transitive. Thus it is absent. In every case ρ_j fixes a_h.

All surviving generators therefore fix $z = a_h$. They contain the pure symmetric groups on $A \setminus \{z\}$ and B. Multiplying ρ_j by its internal B-restriction gives the single crossing transposition, since its only action on A is the crossing. These generate the full symmetric group on $W \setminus \{z\}$, proving the final equality. □

Attribution and scope. The complete anchored-path theorem, its exhausted symmetric images, and the local crossing orbit bound are acquired. The short-action centralizer argument and the exact first-omission analysis are the new steps. This is a general equality theorem with both sides of length at least two. It makes no claim about the remaining unequal-end separator, the short side cases, or actual attachments by outside generators. No original, question, numerical or formalization credit is claimed.